GUJARAT TECHNOLOGICAL UNIVERSITY

Diploma in Engineering — Modern Physics (DI01000061)
Unit 1: Units and Measurements

Topic 1: Physical Quantities

• **Definition:** A physical quantity is any property of an object or material that can be measured using a number and a specific standard unit.

• **3 Features:**

- It can always be measured accurately using tools or instruments.
- It has two main parts: a number and a unit (like 5 kilograms).
- It is divided into basic (fundamental) and combined (derived) types.

• **3 Uses:**

- Helps engineers design machine parts with exact dimensions.
- Allows technicians to calculate weight, speed, and force accurately.
- Ensures manufacturing standards remain the same in every factory.

• **Real-Life Application:** When you measure the exact length of an electrical wire using a ruler before cutting it for a circuit board.

Topic 2: Systems of Units

• **Definition:** A system of units is a complete, standard set of rules used to measure all physical quantities around the world.

• **3 Features:**

- Includes older standard systems like CGS (Centimeter-Gram-Second) and MKS (Meter-Kilogram-Second).
- Uses the modern SI system as the globally accepted standard for engineering.
- Allows simple conversion of values from one system to another.

• **3 Uses:**

- Helps in global trade by keeping measurements identical in all countries.

- Simplifies engineering calculations in textbooks and practical labs.
- Prevents costly errors when different engineering teams share data.

• **Real-Life Application:** Buying fuel in liters or measuring a mechanical component in millimeters, ensuring that imported machine parts fit together perfectly.

Topic 3: Fundamental and Derived Quantities

• **Definition: Fundamental quantities** are basic physical quantities that do not depend on any other quantity to be measured, while **derived quantities** are formed by combining two or more fundamental quantities.

• **3 Features:**

- Fundamental quantities are limited (like length, mass, and time) and act as the foundation.
- Derived quantities are unlimited and created using mathematical formulas (like speed = length ÷ time).
- Both rely on standardized units to express their measured values accurately.

• **3 Uses:**

- Defines the basic building blocks for all physics and engineering formulas.
- Helps calculate complex engineering properties like volume, pressure, and electrical resistance.
- Simplifies dimensional analysis to check if an equation is correct.

• **Real-Life Application:** The mass of a car is a fundamental quantity, but calculating its density or speed to see how it moves involves derived quantities.

Topic 4: Vernier Calipers and Micrometer Screw Gauge

• **Definition:** These are **precision measuring instruments** used by engineers to measure very small lengths, diameters, and thicknesses with high accuracy down to fractions of a millimeter.

• **3 Features:**

- A Vernier caliper measures internal, external, and depth dimensions using a sliding scale.
- A Micrometer screw gauge uses a rotating screw to measure ultra-thin objects like wires.
- Both rely on a "least count," which is the smallest value the instrument can accurately read.

• **3 Uses:**

- Measuring the exact thickness of sheet metal in manufacturing.
- Checking the inner diameter of a pipe to ensure a tight connection.

- Measuring the thickness of thin electrical wires before installation.

• **Real-Life Application:** Using a micrometer screw gauge to check the diameter of an engine piston wire to ensure it perfectly fits inside a cylinder block.

Topic 5: Measurement Errors

• **Definition:** A **measurement error** is the numerical difference between the true value of a physical quantity and the value obtained by measuring it with an instrument.

• **3 Features:**

- Divided into types like absolute error (actual difference), relative error, and percentage error.
- Can happen due to faulty instruments, human mistakes, or environmental changes.
- Can be minimized by taking multiple readings and calculating their average.

• **3 Uses:**

- Assists engineers in understanding the tolerance limits of a design.
- Helps in identifying whether a measuring tool needs recalibration.
- Ensures safety by accounting for structural uncertainties in construction.

• **Real-Life Application:** If a metal rod is exactly 10.0 cm long but your tool reads 9.9 cm, the absolute error is 0.1 cm, which tells you how much your tool is off.